

Appendix B

THE RADIATION FROM AN ACCELERATED CHARGE

Here we give a largely qualitative view of the classical theory of emission of electromagnetic radiation from an accelerated charge, restricting ourselves to the case of a stationary charge in vacuum that is suddenly accelerated to a non-relativistic velocity $v \ll c$.

We know that a stationary charge has an associated static electric field $\mathbf{E}$ whose energy per unit volume is given by

$$\rho = \frac{1}{2} \epsilon_0 E^2 \quad (\text{B-1})$$

This energy is stored in the field and is not radiated away. If the charge moves with a uniform velocity, there is a magnetic field $\mathbf{B}$ associated with it as well as an electric field. The total energy stored in the nonstatic field of a uniformly moving charge is larger than for the static field of a stationary charge, the additional energy being supplied from the work done by the forces that initially produced the motion of the charge. The energy density in this case is given by

$$\rho = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2\mu_0} B^2 \quad (\text{B-2})$$

and the energy stored in the field moves along with the charge. That the energy is not radiated away, even in this case, follows from transforming to a reference frame in which the charge is stationary and applying the relativistic requirement that the behavior of the charge, including whether or not it radiates, cannot depend on the frame of reference from which it is viewed. Hence for a charge having constant velocity, the electric and magnetic fields are able to adjust themselves in such a way that no energy is radiated, even though these fields are not static.

For an accelerated charge, however, the nonstatic electric and magnetic fields cannot adjust themselves in such a way that none of the stored energy is radiated. We can understand this qualitatively by considering the behavior of the electric field. In Figure B-1 we describe this field by drawing some of the lines of force surrounding a charge which was at rest at the initial instant t , suffered a constant acceleration a to the right during the interval t to t' , and then continued moving with a constant final velocity. The figure shows the lines of force at some later instant t'' , as viewed from the frame of reference moving at that velocity v . At small distances the lines of force are directed radially outward from the present position of the charge. At large distances they emanate from where the field would anticipate it to be if unaccelerated. The reason is that information concerning the position of the charge cannot be transmitted to distant locations with infinite velocity, but only with the velocity c . As a result, there are kinks in the lines of force found between a sphere centered on the anticipated position and of radius $c(t'' - t)$, which is the minimum distance at which the field can "know" the acceleration started, and a sphere centered on the actual position and of radius $c(t'' - t')$, which is the minimum distance at which the field can know that the acceleration stopped. As t'' increases, the region containing the kinks expands outward with velocity c . That is, each kink of adjustment propagates along its line of force in much the same way as a kink set up at one end of a long stretched rope propagates along the rope. The electric field in the region containing kinks has components which are both longitudinal and transverse to the direction of expansion. But, by constructing diagrams for several values of t'' , it is easy to see that the longitudinal

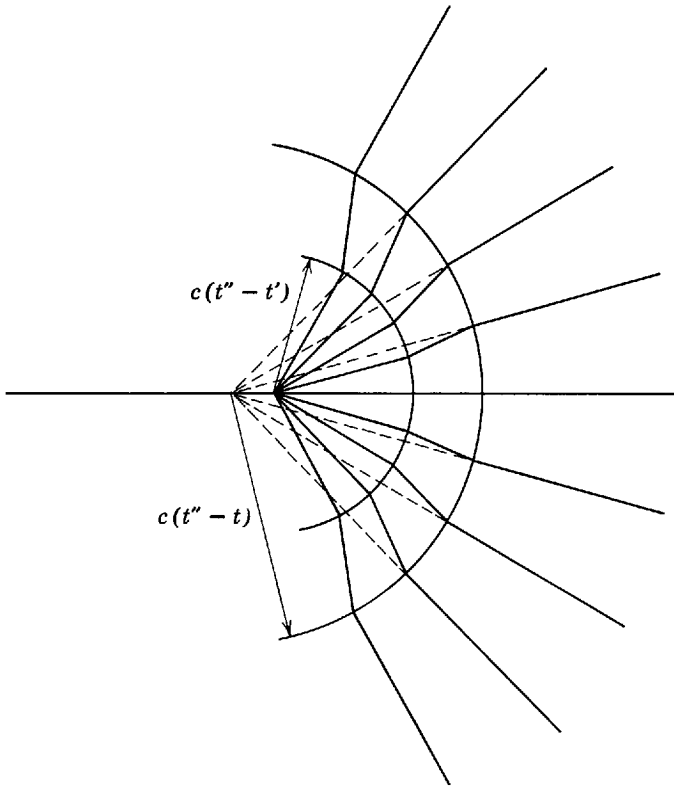


Figure B-1 The lines of force surrounding an accelerated charge. Only some of the lines are shown.

component dies out very rapidly and can soon be ignored, whereas the transverse component dies out slowly. In fact, electromagnetic theory shows, by calculations based upon the same idea as in our qualitative discussion, that at large distances from the region of the acceleration (large t'') the transverse electric field obeys the equation

$$E_{\perp} = \frac{qa}{4\pi\epsilon_0 c^2 r} \sin \theta \quad (\text{B-3})$$

In this equation, which is valid only if $v/c \ll 1$, $r = c(t'' - t)$ is the magnitude of the vector $\mathbf{r}$ from the region at which the acceleration $\mathbf{a}$ took place to the point at which the transverse field is evaluated, and θ is the angle between $\mathbf{r}$ and $\mathbf{a}$. The dependence of $E_{\perp}$ on θ and r can be seen from Figure B-1 and comparable diagrams for larger values of t'' , and it should be clear from our discussion that $E_{\perp}$ must be proportional to q and a . Similarly, there is a transverse magnetic field moving along with $E_{\perp}$, and at large distances from the region of acceleration its strength, if $v/c \ll 1$, is given by

$$B_{\perp} = \frac{\mu_0 qa}{4\pi cr} \sin \theta \quad (\text{B-4})$$

These two transverse fields propagating outward with velocity c form the electromagnetic radiation emitted by the accelerated charge. The radiated field is polarized with $\mathbf{E}$ in the plane of $\mathbf{a}$ and $\mathbf{r}$ and with $\mathbf{B}$ at right angles to this plane. The energy density of the radiation is

$$\rho = \frac{1}{2} \epsilon_0 E_{\perp}^2 + \frac{1}{2} \frac{B_{\perp}^2}{\mu_0}$$

or, with $c = 1/\sqrt{\mu_0 \epsilon_0}$ and $B_{\perp} = E_{\perp}/c$

$$\rho = \frac{1}{2} \epsilon_0 E_{\perp}^2 + \frac{1}{2} \epsilon_0 E_{\perp}^2 = \epsilon_0 E_{\perp}^2 \quad (\text{B-5})$$

The "Poynting vector," which gives the energy flow per unit area (i.e., the intensity of radiation) is directed along $\mathbf{r}$ and has a magnitude

$$S = \rho c = \epsilon_0 c E_{\perp}^2$$

Hence, from (B-3)

$$S = \frac{q^2 a^2}{16\pi^2 \epsilon_0 c^3 r^2} \sin^2 \theta \quad (\text{B-6})$$

which can also be obtained from the relation defining the Poynting vector

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}$$

Notice that no energy is emitted forward or backward along the direction of acceleration ($\theta = 0^\circ$ or 180°) and that the energy emitted is a maximum at right angles to this direction ($\theta = 90^\circ$ or 270°). The radiated energy is distributed symmetrically about the line of accelerated motion and with respect to the forward and backward directions. We see also from (B-6) that the radiated intensity obeys the familiar inverse square law, $S \propto 1/r^2$. To get the rate R at which total energy is radiated in all directions per unit time, i.e., the power, we integrate S over the area of a sphere of arbitrary radius r . That is

$$R = \int S(\theta) dA = \int_0^\pi S(\theta) 2\pi r^2 \sin \theta d\theta$$

in which $dA = 2\pi r^2 \sin \theta d\theta$ is the differential ring-shaped element of area on the sphere in a range between θ and $\theta + d\theta$. Carrying out the integration yields

$$R = \frac{1}{4\pi\epsilon_0} \frac{2}{3} \frac{q^2 a^2}{c^3} \quad (\text{B-7})$$

which is the rate of radiation of energy from the accelerated charge. The rate of radiation is seen to be proportional to the square of the acceleration.

It should be pointed out that energy must be supplied to maintain a constant linear acceleration of the charge, some of it simply to compensate for the energy radiated away. However, the radiation loss is usually negligible at nonrelativistic speeds. In the case of deceleration the radiated energy is supplied by the energy stored in the electromagnetic field of the charge whose velocity is decreasing. This is the bremsstrahlung radiation discussed in Chapter 2.

A frequent application of (B-7) is to a vibrating electric dipole. Let a charge q be vibrating about the origin of the x axis with simple harmonic motion. Then the displacement of the charge as a function of time is $x = A \sin \omega t$ where A is the amplitude of the vibration and $\omega = 2\pi\nu$ its angular frequency. The acceleration of the charge is given by $a = d^2x/dt^2 = -\omega^2 A \sin \omega t = -\omega^2 x$. If we substitute this for a in (B-7) we obtain

$$R = \frac{2q^2\omega^4 x^2}{4\pi\epsilon_0 3c^3} \quad (\text{B-8})$$

Because x varies with time, the power radiated also varies with time at the same frequency as the vibration of the dipole. The average value of $x^2 = A^2 \sin^2 \omega t$ over one period of vibration, however, is simply $A^2/2$, so that the *average* rate of radiation is given by

$$\bar{R} = \frac{q^2\omega^4 A^2}{4\pi\epsilon_0 3c^3}$$

or, with $\omega = 2\pi\nu$

$$\bar{R} = \frac{16\pi^4\nu^4 q^2 A^2}{4\pi\epsilon_0 3c^3} \quad (\text{B-9})$$

Now qx is the electric dipole moment of the vibrating dipole when the charge is at x . So qA is the amplitude of the electric dipole moment. Writing $qA = p$, we have the useful expression

$$\bar{R} = \frac{4\pi^3\nu^4 p^2}{3\epsilon_0 c^3} \quad (\text{B-10})$$

PROBLEM

1. According to the classical electromagnetic theory of Appendix B, what power is radiated by a single electron in a gold atom during the roughly 10^{-12} sec that it takes to collapse from an orbit of radius 1.0×10^{-10} m to the surface of the nucleus, the nuclear radius being about 6.9×10^{-15} m? Assume that all the lost electrostatic energy is radiated, the electron's kinetic energy remaining unchanged during the motion.

